

Headline part X: the stochastic $1/\log N$ regime

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Abstract

Paper-facing wrapper for unit 194: at chart level ($d = 2$, equal starting exponents), for a deterministic corner-vanishing observable amplitude, a positive normaliser amplitude and a common random phase with Taylor data $X_n \Rightarrow Z$, $\log N_n \cdot Z_{N_n}[\phi](X_n)/Z_{N_n}[1](X_n) \Rightarrow B_p^\phi(Z)/A_p^1(Z)$ (`headline_posterior_log_decay_stochastic`), stated on the explicit chart integrals `twoDAMP` with the analytic amplitudes of the Taylor data. This is the stochastic form of `headline_posterior_log_decay`; when $B_p^\phi(Z) = 0$ the limit is 0 and no faster rate is claimed. Zero sorry/axiom.

1 Paper-facing meaning

The empirical posterior expectation of an observable vanishing at the corner of the chart decays like $1/\log N$ with a *random* coefficient determined by the limiting Taylor data of the empirical phase; this closes the “stochastic version of the $1/\log N$ regime” left open in `rem:empirical_expectation_lean` since unit 169.

2 Lean notes

The wrapper is a one-line `exact`: `chartZ` unfolds definitionally to the `twoDAMP` of the analytic amplitude of the coefficient pair.